

Session 6



Grouping and summarizing

The summarize verb

`mean()` and `sum()` are only two of the functions R provides for summarizing a collection of numbers. Here, you'll learn to use the `median()` function in combination with `summarize()`.

```
library(gapminder)
library(dplyr)

gapminder %>%
  summarize(medianLifeExp = median(lifeExp))
```

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**Summarizing the median life expectancy
in 1957**

The group_by verb

Most data operations are done on groups defined by variables. `group_by()` takes an existing data frame and converts it into a grouped data frame where operations are performed "by group".

```
gapminder %>%  
  group_by(year) %>%  
  summarize(medianLifeExp = median(lifeExp), maxGdpPercap = max(gdpPercap))
```

Exercise

Summarizing by continent and year

Instead of grouping just by year, or just by continent, you'll now group by both continent and year to summarize within each.

Find the median life expectancy (`lifeExp`) and maximum GDP per capita (`gdpPercap`) within each combination of continent and year, saving them into `medianLifeExp` and `maxGdpPercap`, respectively.



Visualizing summarized data



Types of visualizations

Line plots

A line plot is useful for visualizing trends over time. In this exercise, you'll examine how the median GDP per capita has changed over time.

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**Visualizing median GDP per capita by
continent over time**

Bar plots

A bar plot is useful for visualizing summary statistics, such as the median GDP in each continent.



Visualizing GDP per capita by country in Oceania

Histograms

A histogram is useful for examining the distribution of a numeric variable. In this exercise, you'll create a histogram showing the distribution of country populations (by millions) in the year 1952.



**Visualizing population with x-axis on a
log scale**



Break
15 minutes

Boxplots

A boxplot is useful for comparing a distribution of values across several groups. In this exercise, you'll examine the distribution of GDP per capita by continent. Since GDP per capita varies across several orders of magnitude, you'll need to put the y-axis on a log scale.

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Adding a title to your graph

Exercise

The `chickwts` dataset will be used, which contains information on the `weight` of chickens fed with different types of `feed`.

- Load the `chickwts` dataset using the `data()` function.
- Create a histogram to represent the distribution of the `weight` variable (chicken weight).
- Create a boxplot to compare the distribution of `weight` across the different types of `feed`.



Data Manipulation with dplyr

Transforming Data with dplyr

Select, mutate, filter, and arrange

In this exercise, you'll put together everything you've learned (`select()`, `mutate()`, `filter()` and `arrange()`), to find the counties with the highest proportion of men.

- Select only the columns `state`, `county`, `population`, `men` and `women`.
- Add a variable `proportion_men` with the fraction of the county's population made up of men.
- Filter for counties with a population of at least ten thousand (`10000`).
- Arrange counties in descending order of their proportion of men.

Aggregating Data

The count verb

The `counties` dataset contains columns for `region`, `state`, `population` and the number of citizens, which we selected and saved as the `counties_selected` table. In this exercise, you'll focus on the `region` column.

```
counties_selected <- counties %>%  
  count(region, sort=TRUE)
```



Counting citizens by state

Mutating and counting

You can combine multiple verbs together to answer increasingly complicated questions of your data. For example: "What are the US states where the most people walk to work?"

You'll use the walk column, which offers a percentage of people in each county that walk to work, to add a new column and count based on it.

The group by, summarize and ungroup verbs

Summarizing

The `summarize()` verb is very useful for collapsing a large dataset into a single observation.

Summarize the `counties` dataset to find the following columns: `min_population` (with the smallest population), `max_unemployment` (with the maximum unemployment), and `average_income` (with the mean of the income variable).

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Summarizing by state



Summarizing by state and region

The top_n verb

The *walk* column offers a percentage of people in each county that walk to work, this column will use to find the total number of people who walk to work in each county.

Now, you'll find the county within each region with the highest percentage of citizens who walk to work.



**Finding the highest-income state in each
region**

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Using `group_by`, `summarize`, `top_n`,
`ungroup` and `count` together

- For each combination of `state` and `metro`, find the total population as `total_pop`.
- Extract the most populated row from each state, which will be either `Metro` or `Nonmetro`.
- Ungroup, then count how often `Metro` or `Nonmetro` appears to see how many states have more people living in those areas.